Basic Introductory CS Course

1. **Introduction:**

   This introductory course will span approximately 8 – 10 weeks. The marking scheme will be TA – MST – ESE :: 20 – 30 – 50. Scoring at least 60% marks will be considered as successful completion.

   There will be 4 courses under this BICS course :

1. Introduction to Computer Science
2. Programming fundamental with C++
3. Basics of Web Development
4. Mathematical Thinking

The timeline for assessment will be:

TA – continuous evaluation (Assignments, Test/Quiz, Project)

MST – after 5 weeks or half or more than half of syllabus completion

ESE – after 10 weeks or complete syllabus completion.

Join these Google Classrooms for specific updates:

1. Introduction to Computer Science
2. Programming fundamental with C++
3. Basics of Web Development
4. Mathematical Thinking

The detailed syllabus is provided below …

Course 1: Introduction to Computer Science

Objective: Build a strong foundation in computer systems, software, networking, and modern computing concepts.

Module	Topics Covered
Module 1: Foundations of Computing	History of Computers, Generations of Computers, Characteristics of Computers, Types of Computers, Computer Applications, Data vs Information, Number Systems (Binary, Decimal, Hexadecimal), Units of Storage, Computer Architecture vs Computer Organization, Hardware vs Software, Types of Software, Open Source vs Proprietary Software
Module 2: Computer Hardware & System Components	Functional Components of a Computer, CPU, ALU, Control Unit, Registers, Cache Memory, Primary Memory (RAM, ROM), Secondary Storage (HDD, SSD), Motherboard, Power Supply, Input Devices, Output Devices, Ports & Connectors, Peripheral Devices
Module 3: Operating Systems Fundamentals	Introduction to Operating Systems, Functions of an OS, Types of Operating Systems, Processes vs Programs, Threads (Basics), Memory Management, File Systems, Process Scheduling (Conceptual), Virtual Memory, Boot Process, Command Line Basics, Linux vs Windows
Module 4: Software Development & Networking	Problem Solving, Algorithms, Flowcharts, Pseudocode, SDLC, Compiler vs Interpreter, Version Control (Git Basics), Computer Networks, LAN, WAN, Internet, IP Address, DNS, HTTP vs HTTPS, Client-Server Architecture, Cloud Computing Basics
Module 5: Modern Computing & Career Paths	Databases, Artificial Intelligence, Machine Learning, Data Science, Cybersecurity Basics, Cryptography Basics, Blockchain, Internet of Things (IoT), Mobile Development, Web Development Overview, Software Engineering Roles, Career Opportunities in Computer Science

Course 2: Programming Fundamentals with C++

Objective: Develop programming logic and object-oriented programming skills using C++.

Module	**Topics Covered**
Module 1: Introduction to Programming	Structure of a C++ Program, Compilation Process, Variables, Data Types, Constants, Input & Output, Operators, Expressions, Type Casting, Keywords, Identifiers, Comments, Debugging Basics
Module 2: Decision Making & Iteration	if Statements, if-else, Nested if, switch, while Loop, do-while Loop, for Loop, Nested Loops, break, continue, Variable Scope, Pattern Printing, Basic Problem Solving
Module 3: Functions, Arrays & Strings	Functions, Function Parameters, Return Values, Function Overloading, Default Arguments, Recursion, Arrays, Multidimensional Arrays, Character Arrays, Strings, Searching Algorithms, Sorting Algorithms, Introduction to Time Complexity
Module 4: Memory Management & Pointers	Memory Layout, Addresses, Pointers, Pointer Arithmetic, Arrays & Pointers, References, Dynamic Memory Allocation, new Operator, delete Operator, Null Pointer, Dangling Pointer, Pointer Applications
Module 5: Object-Oriented Programming	Classes, Objects, Access Specifiers, Constructors, Destructor, this Pointer, Static Members, Friend Functions, Encapsulation, Abstraction, Basic Inheritance, Function Overriding, Introduction to Polymorphism

Course 3: Basics of Web Development

Objective: Learn how websites work and build responsive web pages using HTML, CSS, and JavaScript.

Module	**Topics Covered**
Module 1: Web & HTML Fundamentals	How the Internet Works, Client-Server Architecture, Browsers, HTTP Basics, HTML Document Structure, Head & Body, Headings, Paragraphs, Lists, Tables, Hyperlinks, Images, Audio, Video, Semantic HTML
Module 2: Forms & Advanced HTML	Forms, Input Types, Labels, Buttons, Form Validation, Fieldsets, Meta Tags, SEO Basics, Accessibility, HTML Best Practices, Favicon, Embedding External Resources
Module 3: CSS Fundamentals	CSS Syntax, Selectors, Colors, Units, Box Model, Margin, Padding, Borders, Backgrounds, Typography, Display Property, Positioning, Overflow, Z-index
Module 4: Responsive Web Design	Flexbox, CSS Grid, Media Queries, Responsive Images, Mobile-First Design, CSS Variables, Pseudo Classes, Pseudo Elements, Transitions, Animations, Basic UI Design Principles
Module 5: JavaScript Essentials	Introduction to JavaScript, Variables, Data Types, Operators, Conditional Statements, Loops, Functions, Arrays, Objects, DOM Manipulation, Events, Form Validation, Browser Storage (LocalStorage)

Course 4: Mathematical Thinking (Discrete Structures)

Objective: Build logical reasoning and mathematical foundations essential for Computer Science.

Module	Topics Covered
Module 1: Logic & Mathematical Reasoning	Propositions, Logical Connectives, Truth Tables, Tautologies, Contradictions, Logical Equivalence, Predicates, Quantifiers, Direct Proof, Proof by Contradiction, Proof by Contrapositive, Mathematical Induction
Module 2: Sets, Relations & Functions	Sets, Subsets, Power Sets, Cartesian Product, Set Operations, Venn Diagrams, Relations, Properties of , Functions, One-One & Onto Functions, Inverse Functions, Function Composition
Module 3: Counting Techniques & Probability	Basic Counting Principle, Addition Rule, Multiplication Rule, Factorials, Permutations, Combinations, Inclusion-Exclusion Principle, Pigeonhole Principle, Probability Basics, Conditional Probability, Bayes' Theorem (Introduction)
Module 4: Graph Theory Fundamentals	Graph Terminology, Types of Graphs, Degree of a Vertex, Paths, Cycles, Trees, Spanning Trees, Connected Components, Graph Representation, Introduction to DFS, Introduction to BFS, Graph Applications
Module 5: Number Theory & Boolean Algebra	Divisibility, Prime Numbers, Greatest Common Divisor (GCD), Least Common Multiple (LCM), Euclidean Algorithm, Modular Arithmetic, Congruence Relations, Binary Arithmetic, Boolean Algebra, Logic Gates.

By Order Course Instructor